SYSTEMVERILOG / UVM MICRO-ARCHITECTURE SPECIFICATION

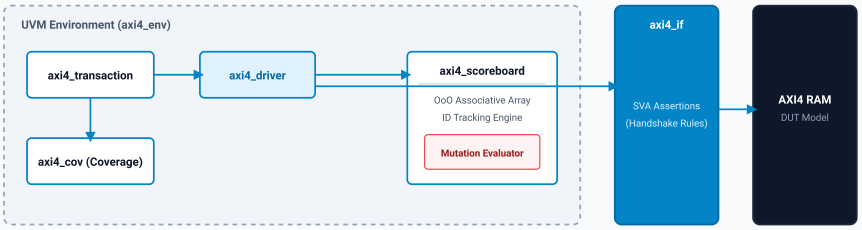
AXI4 Verification IP (VIP) Architecture

Document Type:	Protocol:	Methodology:
Architecture Specification	AMBA AXI4 (IHI0022E)	UVM 1.2 / SVA

1. Architectural Overview

This document defines the structural specification for an AMBA AXI4 Universal Verification Methodology (UVM) Verification IP. The environment validates dynamic read and write channels across split transactions, out-of-order completion, and pipelined burst requests. Integrated SystemVerilog Assertions (SVA) enforce handshake rules directly on the interface, while an associative scoreboard tracks active transaction IDs.

2. Verification IP Flow Diagram



3. Component Functional Specifications

The VIP contains modular SystemVerilog and UVM elements configured for high protocol fidelity:

- **Sequence Item (`axi4_transaction.sv`):** Holds randomized control attributes including address vectors, burst properties (`FIXED`, `INCR`, `WRAP`), dynamic data arrays, and transaction IDs (`AWID`, `ARID`). An embedded `covergroup` measures bin distribution across length combinations.
- **Driver Component (`axi4_driver.sv`):** Pulls sequence items from the sequencer and manages signal timing across physical interface signals on positive clock edges.

3. Component Specifications (Continued)

- **Scoreboard Engine (`axi4_scoreboard.sv`):** Maintains dual associative arrays indexed by `AWID` and `ARID`. This design supports concurrent outstanding transactions and out-of-order responses without losing packet tracking state.
- **Mutation Fault Injection Engine:** Embedded in the verification logic to inject systematic protocol corruptions. This provides quantitative feedback by evaluating whether checkers detect induced anomalies.

4. Protocol Compliance Assertions (SVA)

SystemVerilog Assertions execute concurrently in `axi4_if.sv` to verify handshake timing conditions. Crucially, control signals (`AWVALID`, `ARVALID`, `RVALID`) must remain asserted once set until the target interface returns a corresponding `READY` signal. Dropping valid indicators prematurely triggers error alerts.

5. Verification Quality Metrics

Verification progress and environmental stability are tracked using the following targets:

Metric Category	Implementation Layer	Evaluation Mechanism	Target State
Interface Rules	`axi4_if.sv`	SystemVerilog Assertions (SVA)	**Zero Violations**
Functional Coverage	`axi4_transaction.sv`	Covergroups & Cross Bins	**100% Closure**
Data Integrity	`axi4_scoreboard.sv`	Associative ID Hash Tables	**Zero Mismatches**
Checker Quality	Scoreboard Evaluator	Mutation Testing Fault Engine	**100% Detection**

6. Integration Summary

Connecting the VIP to the `axi4_ram_dut` memory model validates full write and read data loops. The environment provides structural correctness, out-of-order tracking support, and quantitative quality evaluation required for thesis presentation and portfolio review.